

Closed-form reproduction of FCS Property 7 at $N > 2$

Three lenses re-applied to uniform all-to-all inhibition, calibrated against the FCS-LI&F oracle on the (w, N) plane.

Nikan Zandian `deq/closed_form_wta_multi/`

1. Context

The companion thread `deq/closed_form_wta/` [1] reads FCS Property 7 (winner-takes-all stability within 4 ticks) of [2] on the 2-neuron contralateral motif through three lenses — Siegert [3], [4] mean-field fixed-point enumeration, $H(\omega)$ latency gate [5], and quasi-renewal mesoscopic simulation [6] at finite population N_{pop} — calibrated against the FCS-LI&F oracle on a 40×40 grid over the two inhibitory weights (w_{12}, w_{21}) . The three 2-neuron findings:

1. Siegert mean-field tracks the FCS Fig. 10 WTA boundary at Jaccard ≈ 0.80 . The $\approx 20\%$ gap is the **diagonal staircase** of integer-tick synchronous lock that smooth-rate theory cannot resolve.
2. The $H(\omega)$ eigenvalue gate $|\text{Re}(\lambda_{\text{dom}})| > 1/T_{\text{FCS}} = 0.25$ closes part of that gap by removing slow-decay cells near the rate-equation bifurcation.
3. Quasi-renewal at $N_{\text{pop}} \rightarrow \infty$ converges back to the Siegert mean-field, confirming that finite-size noise alone cannot undo the spike-timing lock.

This thread extends the same scenario to **more than two neurons**, sweeping uniform all-to-all lateral inhibition across $N \in \{2, 3, 4, 6, 10\}$ and $w \in \{-40, \dots, -1\}$. The driving question: do the three lenses’ relative predictive ordering carry over from $N = 2$ to $N > 2$?

Spoiler. At $N > 2$ the relationship breaks in two distinct, opposite ways. Both are exhibits of “FCS integer-tick discreteness sees what smooth theory cannot”, but they have opposite signs.

2. Setup

2.1. Topology

Uniform all-to-all inhibition: W has $W_{i,j} = w$ for $i \neq j$ and $W_{i,i} = 0$. Every neuron has its own external drive of magnitude `self_drive` = 11 (the FCS delayer threshold), with neuron 0 receiving an extra integer self-drive bump `drive_bump` $\in \{0, 1\}$ to optionally break the FCS Lustre-faithful S_N permutation symmetry. The constructor `deq/archetypes/topologies.py:all_to_all_inhibition` reduces to `contralateral(w, w)` at $N = 2$, `drive_bump` = 0 (bit-exact unit-checked).

2.2. Gates

The Phase 0 oracle uses two gates side by side:

1. **Strict** (FCS Fig. 10 generalization): `post-warmup rate_max` ≥ 0.99 AND `second_max` ≤ 0.01 . Coincides with the 2-neuron gate at $N = 2$.
2. **Margin** (aligns with Phases 1, 3): `rate_max` $-$ `second_max` ≥ 0.30 .

Phases 1, 2, 3 use the **margin gate** for cross-comparison.

3. Phase 0 — FCS oracle at $N > 2$



Figure 1: Phase 0 FCS oracle labels across $(w, N, \text{drive_bump})$. Left column: strict gate; right: margin gate. Per panel: two rows of dots for $\text{drive_bump} \in \{0, 1\}$.

3.1. Headline counts ($\text{drive_bump} = 1$)

N	Strict	Margin	Comment
2	9 / 40	10 / 40	Bumped diagonal cuts the 2-neuron staircase.
3	4 / 40	9 / 40	Near-WTA cells (winner ≈ 0.98 , second > 0.01).
4	2 / 40	6 / 40	Integer-tick lock collapses into period- N round-robin.
6	0 / 40	8 / 40	Strict gate fails; margin recovers a pocket.
10	0 / 40	3 / 40	Thin band at $w \in \{-15, -16, -7\}$ only.

At $\text{drive_bump} = 0$ every cell is red at every N . The FCS Lustre-faithful S_N symmetry forces all N neurons into identical synchronous trajectories. **The 2-neuron diagonal staircase becomes an N -way synchronous lock at every N .**

4. Phase 1 — Siebert FP enumeration (multi-restart fsolve)

The 2-neuron 1-D scalar reduction does not extend past $N = 2$. We initially attempted a **symmetric-orbit ansatz** under S_N (orbits of shape (ν_W^k, ν_L^{N-k}) , exact 2-D system per k), but the multi-restart fsolve diagnostic flagged that the broken-symmetry S_{N-1} case at $\text{drive_bump} = 1$ admits FPs with **three or more distinct rates**. For instance at $N = 3$, $w = -30$, $\text{drive_bump} = 1$, fsolve from random ICs returns the FP $[0.23, 0.12, 0.03]$, which the “bumped + $(N - 1)$ -tied losers” ansatz cannot represent.

Per the plan’s risk-flag, Phase 1 falls back to **multi-restart fsolve in the full N -dimensional rate vector**. The IC set is 30 per cell: corner ICs (symmetric, k -winners- $(N - k)$ -losers for $k = 1, \dots, 4$, extreme and mild variants) plus random fills. WTA-capable iff some FP has $\text{rate_max} - \text{rate_2nd} \geq 0.30$.

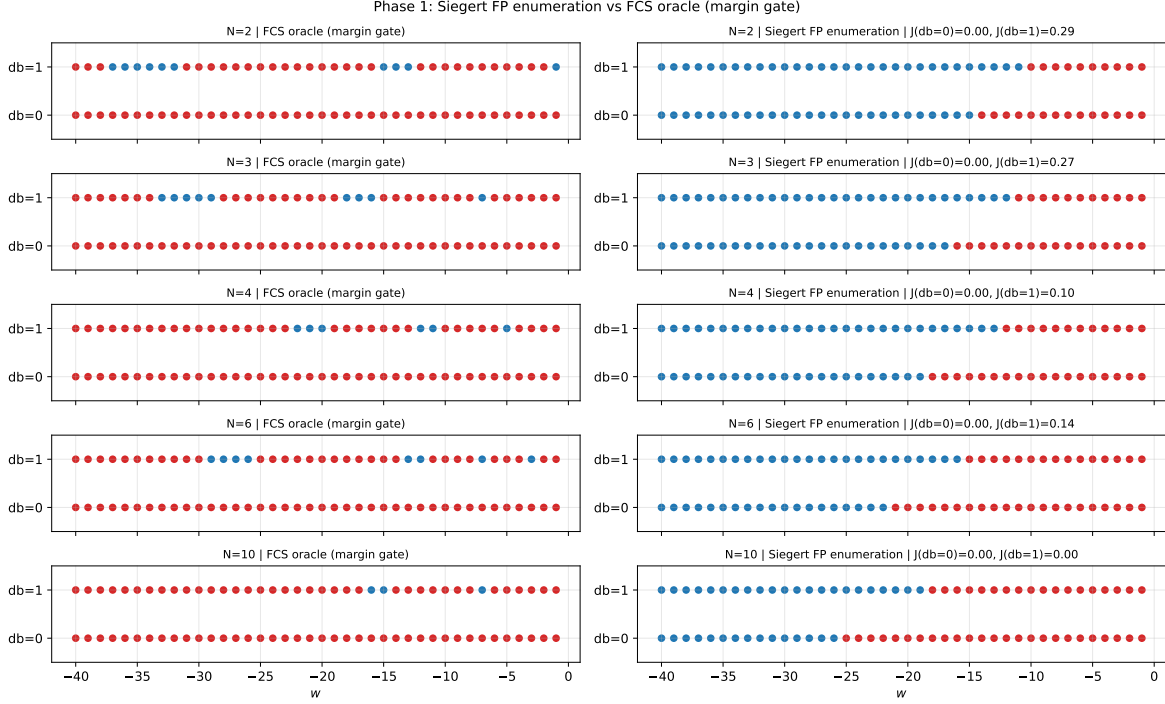


Figure 2: Phase 1 Siebert FP enumeration (right) vs FCS margin gate (left), per N , per drive_bump . At every N , $J_{\text{db}=0} = 0$: Siebert sees bistable FPs that the symmetric FCS dynamics cannot select.

4.1. Two distinct invisibility phenomena

1. **drive_bump = 0: staircase invisibility generalizes to N -way lock.** Siebert sees the bistable asymmetric FP; FCS sees synchronous lock. FCS-blue = 0 everywhere; Siebert-blue large. Direct generalization of the 2-neuron staircase.
2. **drive_bump = 1, high N : inverse staircase.** At $N = 10$, the w values where FCS sees clean WTA ($w \in \{-15, -16, -7\}$) admit **no asymmetric rate-equation FP at all**. 270 curated initial conditions all converge to the unique symmetric FP $(0.13, 0.046, \dots, 0.046)$ with spread $0.09 < 0.30$. The FCS integer-tick dynamics produce WTA **below** the smooth-rate bifurcation threshold.

This second phenomenon is the **opposite** of the 2-neuron staircase. At $N = 2$, FCS sees synchronous lock that Siebert overpredicts away (FCS-red, Siebert-blue). At $N = 10$, FCS sees integer-tick WTA that Siebert under-predicts (FCS-blue, Siebert-red). Recall (Siebert $\cap$ FCS / FCS) at $\text{drive_bump} = 1$:

N	FCS-blue (margin)	Recall
2	10	0.90
3	9	0.89
4	6	0.50
6	8	0.50
10	3	0.00

Recall degrades monotonically with N , collapsing to zero at $N = 10$.

5. Phase 2 — $H(\omega)$ latency gate

The $N \times N$ rate-equation Jacobian at the $k = 1$ winner FP has a **clean block structure** under the S_{N-1} loser-permutation subgroup. The numerics at $\text{drive_bump} = 1$, $w = -30$ show an $(N - 2)$ -fold degenerate “loser-permutation” eigenvalue locked at -0.424 in Siegert units at every $N \geq 3$; the two non-degenerate eigenvalues are the winner-loser-mean 2-block.

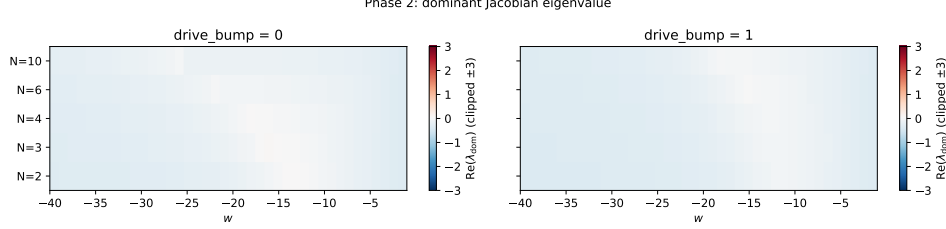


Figure 3: Dominant Jacobian eigenvalue $\text{Re}(\lambda_{\text{dom}})$ across (w, N) . For $|w| \geq 19$ the asymmetric FP is stable at every N ; the gate $|\text{Re}(\lambda_{\text{dom}})| > 0.25$ passes there.

The gate, however, **cannot rescue** Phase 1’s two failure modes. It evaluates eigenvalues at the **existing** rate-equation FP — where that FP does not exist (inverse staircase) or is not selected (staircase), no downstream gate helps. Overall $J(H, \text{FCS-margin}, \text{db}=1) = 0.149$ vs $J(\text{Siegert}) = 0.164$: the gate slightly tightens Siegert by removing some slow-decay cells but does not improve agreement with FCS.

6. Phase 3 — Quasi-renewal at finite N_{pop}

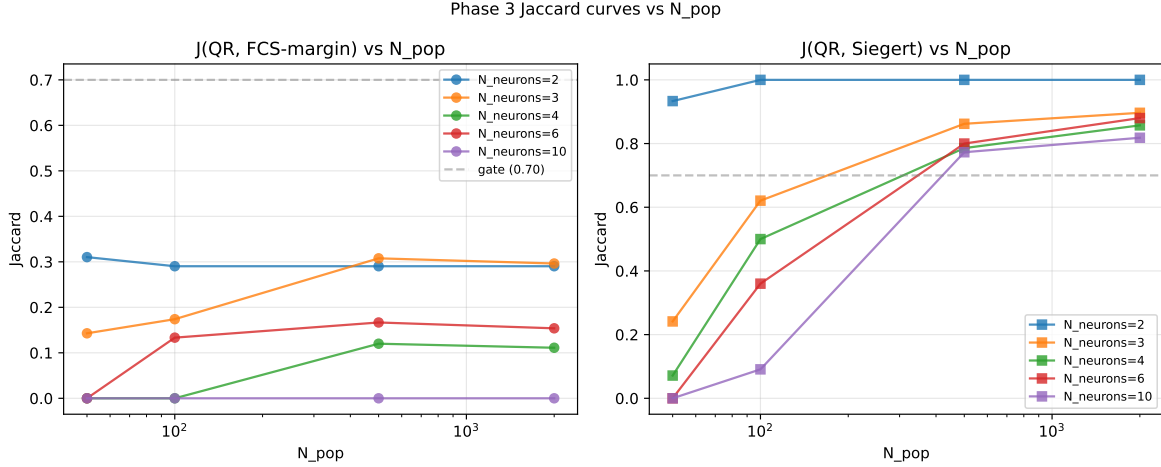


Figure 4: Left: Jaccard(QR, FCS-margin) vs N_{pop} , one curve per N_{neurons} . Right: Jaccard(QR, Siegert). Mean-field convergence is clean at every N_{neurons} .

6.1. Mean-field convergence headline

QR at $N_{\text{pop}} = 2000$ vs Siegert mean-field:

N_{neurons}	$J(\text{QR}, \text{FCS})$	$J(\text{QR}, \text{Siegert})$
2	0.29	1.00
3	0.30	0.90
4	0.11	0.86
6	0.15	0.88
10	0.00	0.82

Mean Jaccard = 0.89, comfortably above the 0.70 gate. The **rate equation is still the right large- N_{pop} object** at every $N_{\text{neurons}} \in [2, 10]$.

Finite- N_{pop} broadening **amplifies with N_{neurons}** : at $N_{\text{neurons}} = 2$ we need $N_{\text{pop}} \approx 50$ for QR to reproduce all Siegert-blue cells; at $N_{\text{neurons}} = 10$ we need $N_{\text{pop}} \geq 500$. Each additional competitor amplifies the noise-induced symmetry restoration of the asymmetric basin.

Crucially, **neither QR nor Siegert recovers the FCS-only inverse-staircase cells** at $N_{\text{neurons}} = 10$ (any N_{pop}). The inverse staircase is genuinely beyond rate-equation theory and its stochastic mesoscopic approximation.

7. Conclusions

The 2-neuron $J \approx 0.80$ Siegert-vs-FCS agreement does **not** extend to $N > 2$. The two phenomena that emerge in its place are:

1. **N -way synchronous lock at drive_bump = 0**: the diagonal staircase of the 2-neuron thread is one slice of a more general S_N integer-tick synchrony at every N . Smooth-rate theory is invisible to it at any N .
2. **Inverse staircase at large N with drive_bump = 1**: a new phenomenon not visible at $N = 2$. The FCS integer-tick dynamics produce clean WTA at specific integer w values **below the smooth-rate bifurcation threshold**. The 1-bumped-neuron, $(N - 1)$ -symmetric-loser configuration is a discrete-arithmetic attractor that has no rate-equation counterpart.

The $H(\omega)$ latency gate cannot bridge either gap because both are properties of the FCS dynamics itself, not of the linearization at any rate-equation FP. The quasi-renewal stochastic mesoscopic equation converges to Siegert mean-field at every N in $[2, 10]$, confirming that the rate equation **is** the right large- N_{pop} object, but inheriting Siegert's blindness to the integer-tick phenomena.

Methodological observation. The 2-neuron's 1-D scalar reduction is a small- N luxury. The natural multi- N tool is the symmetric-orbit ansatz under S_N , exact for uniform W at zero symmetry breaker. With drive_bump > 0 the broken-symmetry structure becomes richer than the ansatz can handle and **multi-restart fsolve in the full N -dimensional rate vector** is the necessary fallback. The plan's risk-flag in this direction proved load-bearing.

8. Pointer to companion thread

For the 2-neuron Property 7 reading and the original three-lens analysis, see `deq/closed_form_wta/note/closed_form_wta_note.pdf` (refs.bib carried forward verbatim).

References

- [1] N. Zandian, "Closed-form reproduction of FCS Property 7 on the 2-neuron contralateral inhibition motif." 2026.
- [2] E. De Maria *et al.*, "On the Use of Formal Methods to Model and Verify Neuronal Archetypes," *Frontiers of Computer Science*, 2020.
- [3] A. J. F. Siegert, "On the First Passage Time Probability Problem," *Physical Review*, vol. 81, no. 4, pp. 617–623, 1951, doi: 10.1103/PhysRev.81.617.
- [4] N. Brunel, "Dynamics of Sparsely Connected Networks of Excitatory and Inhibitory Spiking Neurons," *Journal of Computational Neuroscience*, vol. 8, no. 3, pp. 183–208, 2000, doi: 10.1023/A:1008925309027.

- [5] M. J. E. Richardson, “Firing-rate response of linear and nonlinear integrate-and-fire neurons to modulated current-based and conductance-based synaptic drive,” *Physical Review E*, vol. 76, no. 2, p. 21919, 2007, doi: 10.1103/PhysRevE.76.021919.
- [6] R. Naud and W. Gerstner, “Coding and Decoding with Adapting Neurons: A Population Approach to the Peri-Stimulus Time Histogram,” *PLoS Computational Biology*, vol. 8, no. 10, p. e1002711, 2012, doi: 10.1371/journal.pcbi.1002711.